

Python Programming Assignment-I

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1. Compute the following expressions and print the outputs to the screen. If $x = 7$, $y = 5$, $z = 11$

(i) $x + y$ (ii) $x + 2y$ (iii) $\frac{x}{y}$ (iv) x^2 (v) x^y (vi) $\frac{x^y}{y^x}$ (vii) $\frac{x+y+z^x+\frac{xy}{z}}{z+x^z+xyz}$

Note: You must check your computer outputs by computing above expressions in the Calculator.

2. Type the below code in the editor and fill the blanks in the print statement.

```
# write your roll number.
# date of execution.
x = 2
y = 3
z = 0
# this is comment line
z = x + y
print(z)
print("above operation is_____")
z += x
print ( z)
print("above operation is_____")
z *= x
print (z)
print("above operation is_____")
z /= x
print (z)
print("above operation is_____")
z %= x
print (z)
print("above operation is_____")
z **= x
print (z)
print("above operation is_____")
z //= x
print (z)
print("above operation is_____")
```

Repeat the above expressions for three different choices of x, y and z.

3. Write a program to convert Celsius scale temperature to Fahrenheit scale temperature using the formula

$$F = \frac{9}{5}C + 32$$

Print the corresponding Fahrenheit temp. of $0^\circ C$, $30^\circ C$ and $100^\circ C$.

4. Compute the area of a circle. Radius is the user input from keyboard. you may use math package.

```
from math import pi    # import pi from math package.
```

```
from math import * # import all of the functions from math package.
```

5. Write two programs that will ask the user for positive integers m and n (with mn) as inputs and print out

$$\sum_{i=m}^n \frac{i^2 - 1}{i^4 + 1}$$

Your first program should use an explicit for loop and the second should use list comprehension.

Note: If you want to calculate $\sum_{i=1}^4 i^2$, we can use both the code below,

```
print( sum([i**2 for i in range(1,5)])) # List comprehension method. This one-liner is sufficient
```

another way of doing this

```
s=0.0
for i in range(1,5):
    s=s+i**2
print(s)
```

6. Use a list comprehension to create a list of squares of the numbers between 10 and 20, including the endpoints.
7. For the model used in introductory physics courses, a projectile thrown vertically at some initial velocity v_i has position $y(t) = y_i + v_i t - \frac{1}{2}gt^2$, where $g = 9.8m/s^2$. Write a Python program that creates two lists, one containing time data (50 data points over 5 seconds) and the other containing the corresponding vertical position data for this projectile. The program should ask the user for the initial height y_i and initial velocity v_i , and print $y(t)$ vs. t .
8. Investigate the code:

```
n =1
under = 1.0
over = 1.0
while n <1076:
    under = under/2.
    over = over * 2.
    print(n, under, over)
    n = n + 1
```

What conclusion could you draw from the code?

9. Investigate the code:

```
N = 1075
eps = 1.0
for i in range(N):
    eps = eps/2
    one_plus_eps = 1.0 + eps
    print(i, 'eps =', eps, 'one + eps =', one_plus_eps)
```

Explain the code.

10. Determine whether a given integer number is prime.
11. Construct a list of prime numbers between two given number.
12. Determine the prime factors of a given positive integer.

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Note: In number theory, the prime factors of a positive integer are the prime numbers that divide that integer exactly. The prime factorization of a positive integer is a list of the integer's prime factors, together with their multiplicities; the process of determining these factors is called integer factorization. The fundamental theorem of arithmetic says that every positive integer has a single unique prime factorization.

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13. Determine the greatest common divisor(gcd) of two positive integer numbers.
14. Determine whether two positive integer numbers are coprime. (Two numbers are coprime if their greatest common divisor equals 1)
15. Difference equation for generating the famous Fibonacci numbers: (Fibonacci arrived at this equation when modelling rabbit populations.)

$$x_n = x_{n-1} + x_{n-2}, \quad x_0 = 1, \quad x_1 = 1, \quad n = 2, 3, \dots$$

- (i) write a program to calculate first n Fibonacci numbers. n may take values 10, 100, 1000, etc
- (ii) Compute the limit

$$\lim_{n \rightarrow \infty} \frac{x_n}{x_{n-1}}$$

[Correct up to 4th decimal place.]

16. The Catalan numbers C_n are a sequence of integers 1, 1, 2, 5, 14, 42, 132...that play an important role in quantum mechanics and the theory of disordered systems. (They were central to Eugene Wigners proof of the so-called semicircle law.) They are given by

$$C_0 = 1, \quad C_{n+1} = \frac{4n+2}{n+2} C_n$$

Write a program that prints in increasing order all Catalan numbers less than or equal to one billion.

17. Find the sum of all the multiples of 3 or 5 below 1000.
18. Compute the difference between the sum of the squares of the first n natural numbers and the square of the sum, where $n=500, 1000, 2000$.
19. **Goldbach's conjecture says that every positive even number greater than 2 is the sum of two prime numbers.** It is one of the most famous facts in number theory that has not been proved to be correct in the general case. It has been numerically confirmed up to very large numbers.
 - (i) Write a program to find the two prime numbers that is sum of a given even integer.
 - (ii) Print the list of first n even numbers and their corresponding Goldbach composition. n may be 50, 100, 200, 300.
20. **Pentagonal numbers** are generated by the formula, $P_n = \frac{n(3n-1)}{2}$. The first ten pentagonal numbers are: 1, 5, 12, 22, 35, 51, 70, 92, 117, 145, ...It can be seen that $P_4 + P_7 = 22 + 70 = 92 = P_8$. However, their difference, $70 - 22 = 48$, is not pentagonal.
Find the pair of pentagonal numbers, P_j and P_k , for which their sum and difference are pentagonal and $D = |P_k - P_j|$ is minimised; what is the value of D ?

21. Write a program to calculate $1 + 1/1! + 1/2! + 1/3! + \dots + 1/10!$. Display the sum after each term is added.
22. Write a program to calculate $1^1 + 2^2 + 3^3 + \dots + 10^{10}$.
23. Write a program to sort the data in ascending and descending order by using Insertion and Bubble sort.
24. Write a program to compute the mode, median and mean of the following:
 - (i) 3, 12, 11, 7, 5, 5, 6, 4, 10 .
 - (ii) 16, 19, 10, 24, 19 .
 - (iii) 1.5977, 1.6028, 1.6108, 1.6067, 1.5995, 1.6064, 1.6080, 1.6054, 1.6098, 1.6049, 1.6064, 1.6179, 1.6082, 1.6095, 1.6134 .

Note: **Mean (Arithmetic Mean):** $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

where x_i : data values and n : number of data

The **median** value of a set of data is the middle value of the ordered data.

The **modal** value of a set of data is the most frequently occurring value.

25. A student received an A in Mathematical Physics (3 credits), a C in Mechanics (3 credits), a B in Experimental Physics (4 credits), and a D in Language (2 credits). Assuming $A = 4$ grade points, $B = 3$ grade points, $C = 2$ grade points, $D = 1$ grade point, and $F = 0$ grade points, find the students grade point average.

Age	Frequency
20-29	11
30-39	25
40-49	14
50-59	7
60-69	3

Table 1: Age Distribution

26. Table 1 gives the age distribution of individuals starting new companies. Find the mean, median, and mode for this distribution.

Note: Mean: 38.8, Median: 37.1, Mode : 34.5

Reference:

- A Primer on Scientific Programming with Python: Hans Petter Langtangen.
- Computational With Physics Python: Dr. Eric Ayars.
- Problem Set on Python Programming: A. Dasgupta (IISER-K)
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- https://www3.nd.edu/~newman/phys_30421/reading/ch2_programming.pdf
- <http://www-personal.umich.edu/~mejn/computational-physics/>
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